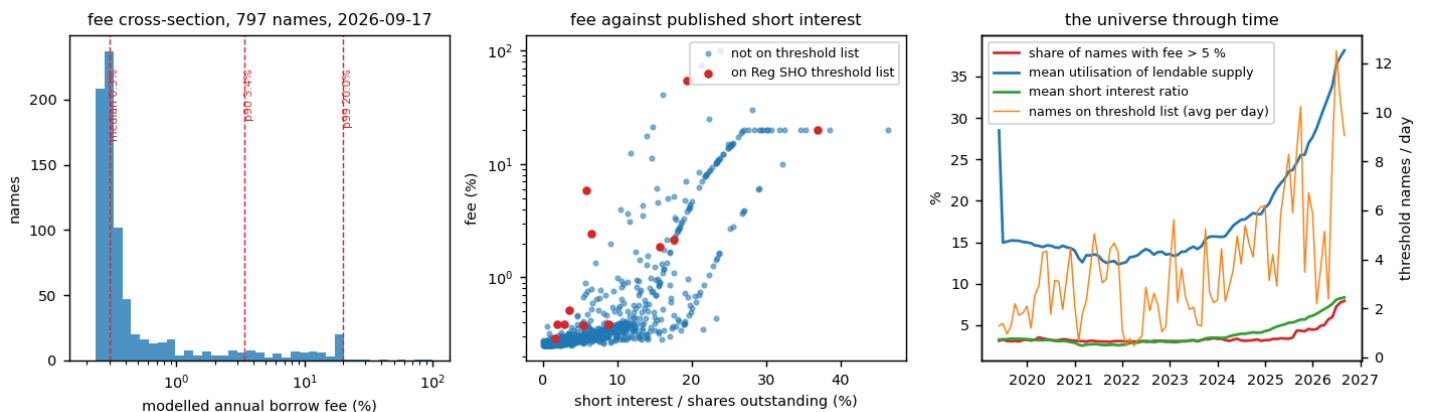


Securities Lending and Financing Analytics for a Long-Short Book

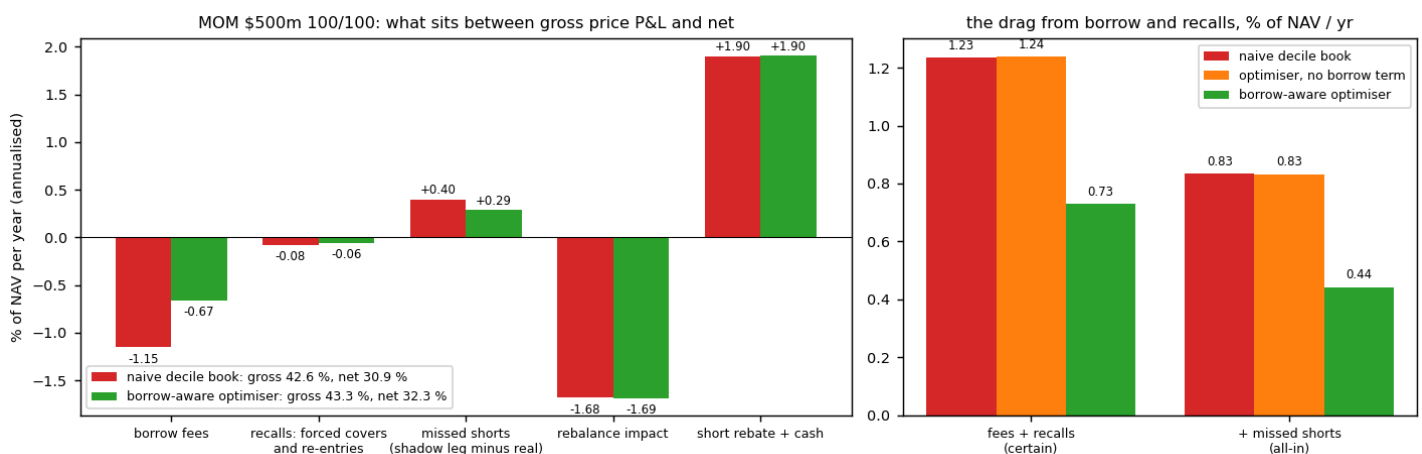
Question. A systematic long-short equity book pays to borrow every share it is short, loses a short when the lender recalls it, and sometimes cannot find the stock at all. How much of the book's return goes to borrow fees and recalls, how often does a short leg become unavailable, and how much of that does a portfolio optimiser that knows the borrow cost get back? This stack answers with a borrow-cost database built from the free public record (FINRA bi-monthly short interest, the Nasdaq and NYSE Regulation SHO threshold lists, the SEC's fails-to-deliver files, a point-in-time ticker map from the CUSIPs in those files, shares outstanding and the effective federal funds rate), a daily simulation of locates, recalls, fees, rebates and margin on three monthly long-short books, a borrow-aware optimiser, and a financing layer with a simulated prime-broker statement reconciled against the book's own accruals.

Method. Python package (slb) on DuckDB and parquet. The borrow model turns what a desk can observe on each day (published short interest and days to cover with FINRA's nine-business-day lag, threshold-list membership from the next morning, fails to deliver a month later, dollar volume and price) into a specialness score, maps the score to an annual fee through the published cross-section of fees (D'Avolio 2002; Engelberg, Reed and Ringgenberg 2018; Muravyev, Pearson and Pollet 2022), with a utilisation floor (Kolasinski, Reed and Ringgenberg 2013), and derives lendable supply, utilisation, a locate probability and a daily recall hazard. Three signals (12-1 momentum, published short interest, one-month reversal) build equal-weighted decile books, 100 % long and 100 % short of a \$500m NAV, rebalanced monthly. Each book runs four ways: with no lending frictions, naive, through the optimiser without the borrow term, and through the borrow-aware optimiser. A shadow short leg holds every intended short so the return the book missed is measured rather than assumed. The financing layer accrues fees, rebates and cash interest ACT/360, computes Regulation T and portfolio margin, and reconciles a prime-broker statement with nine kinds of seeded discrepancy.

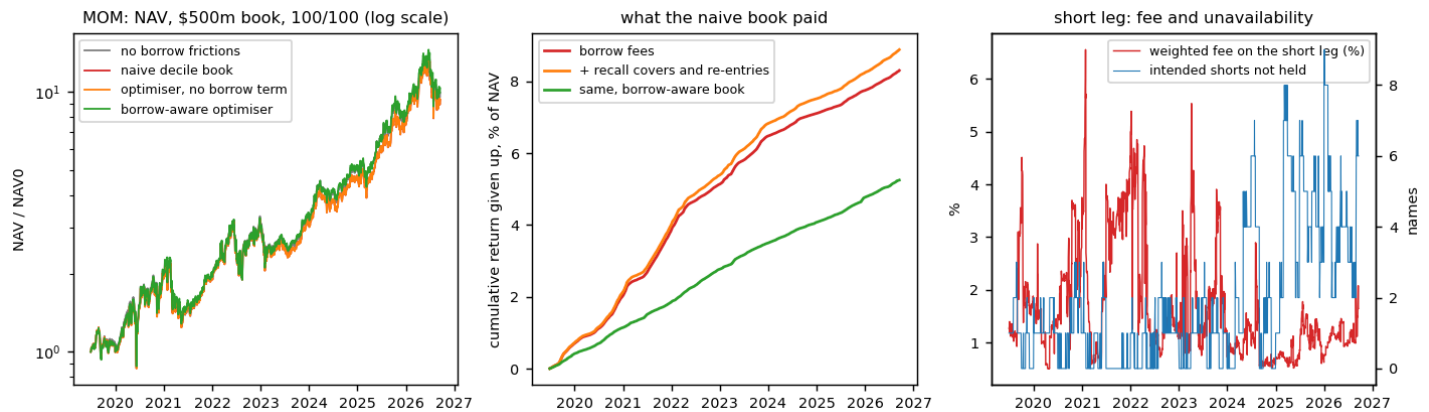
Caveats. Interactive Brokers' shortable file (the one free source of actual fees) could not be reached from this network or from GitHub's runners, so the fees are modelled and the fee level is a stated assumption checked by sensitivity; the code fits the map to IB snapshots when the fetch workflow delivers them. The universe is selected on the latest FINRA file, so it is heavily shorted by construction and survivorship-biased; strategy returns are context, the drag is the object of study.



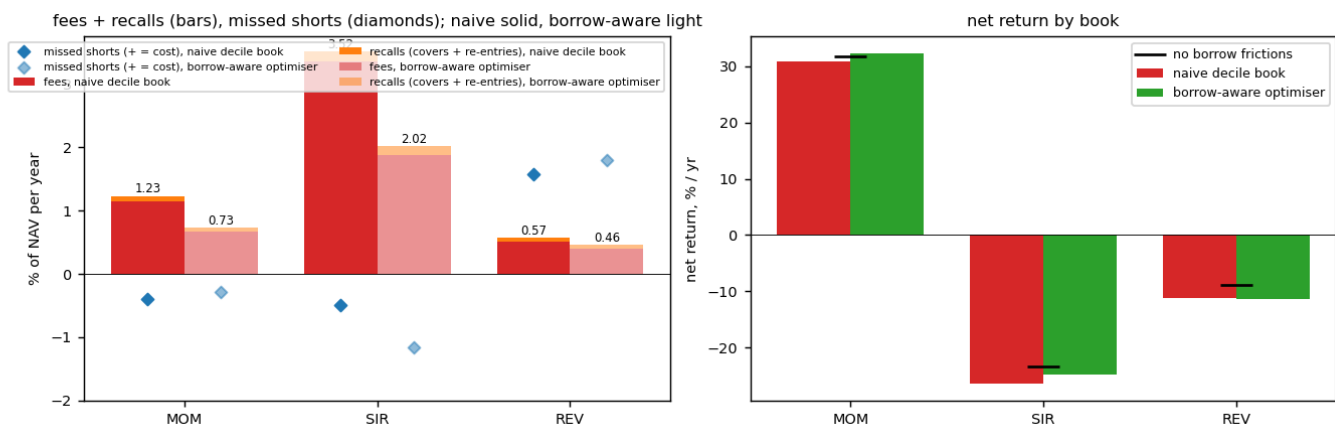
The borrow model: the modelled fee cross-section on the last day against the published quantiles, fee against published short interest with the threshold-list names marked, and the universe's short interest, utilisation and specials through time.



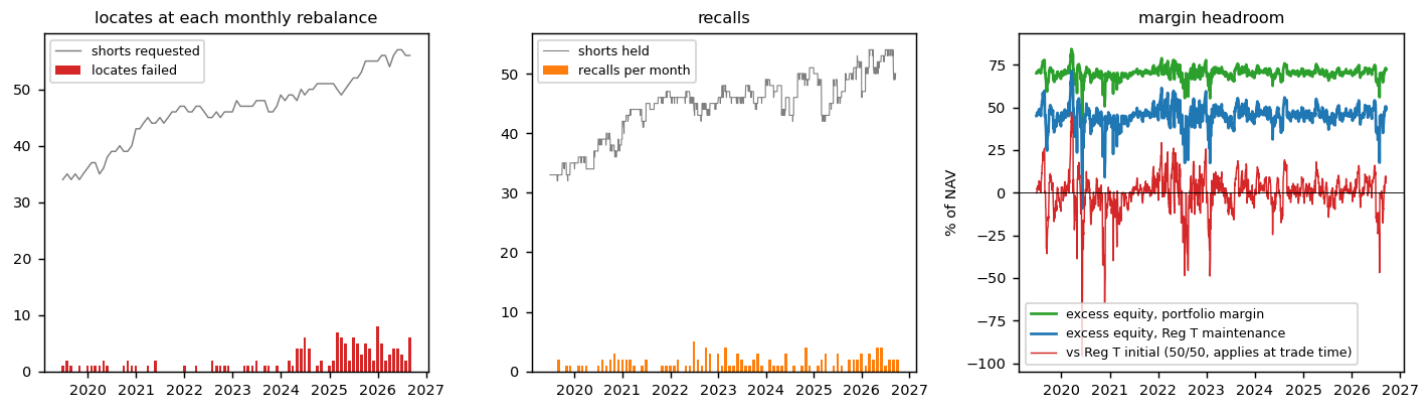
The one number: annualised decomposition of the headline book from gross price P&L to net, naive and borrow-aware, with the drag from borrow and recalls.



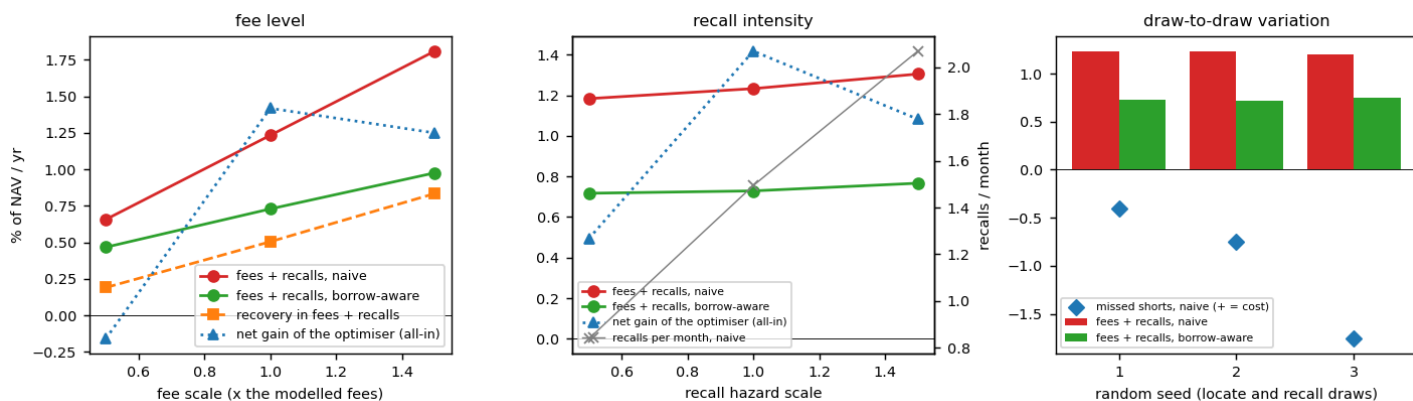
The headline book under the four treatments; what the naive book paid in fees and recalls; the short leg's weighted fee and the intended shorts it could not hold.



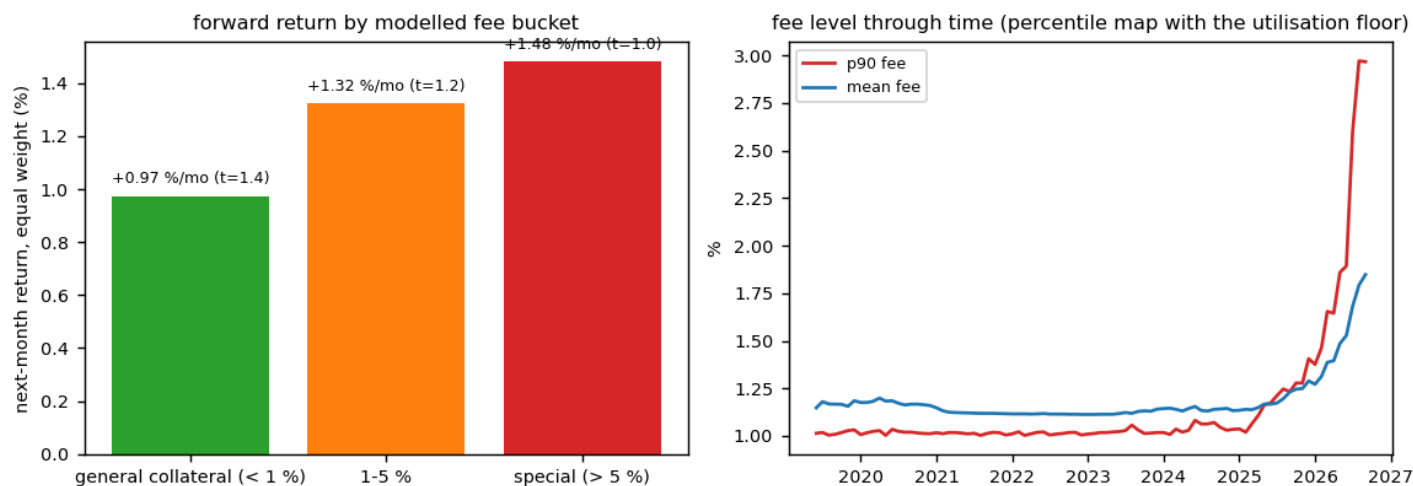
The three books: drag components naive against borrow-aware, and net returns.



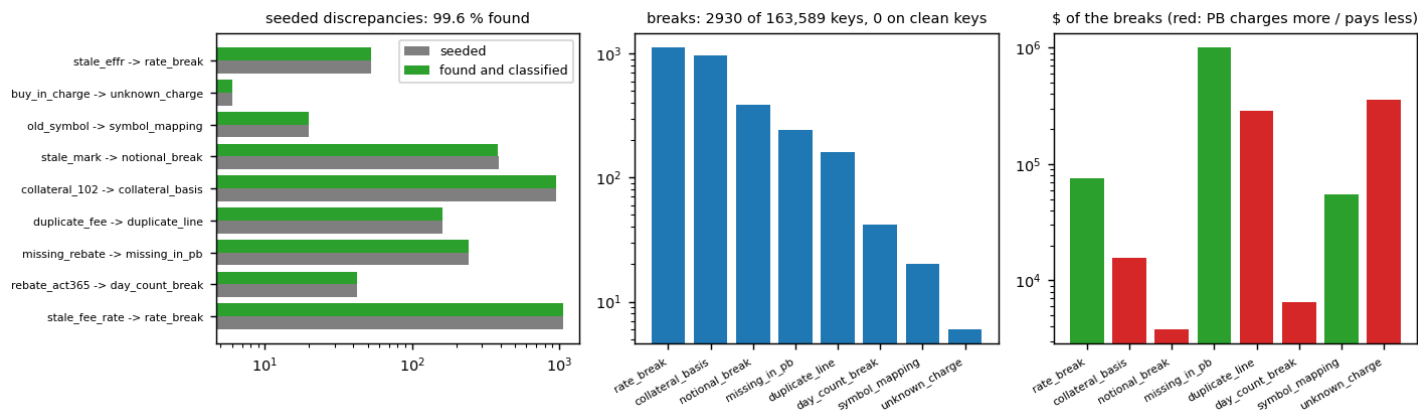
Unavailability: locates failed at each rebalance, recalls per month, and the margin headroom under Regulation T and portfolio margin.



Fee level, recall intensity and random-draw sensitivity of the drag and of the optimiser's recovery.



A check on the fee ordering: next-month returns by fee bucket (expensive-to-short names should underperform), and the fee level through time.



The prime-broker reconciliation: seeded discrepancies found by type, breaks by type and their dollar size.

Generated by scripts/summarize.py from results/run.json.

The one number

On the MOM book (\$500m NAV, 100 % long / 100 % short of NAV, monthly rebalance, 87 rebalances from 2019-06-28 to 2026-08-31, equal-weighted deciles of 801 names), borrow fees and recalls cost 1.23 % of NAV a year (1.15 % in fees at a weighted fee of 1.37 % on the short leg, 0.08 % in forced covers and re-entries: 1.5 recalls a month on about 46 shorts). 3.6 % of the shorts requested at a rebalance could not be located and the book was missing an intended short on 3.8 % of position-days; the price return the book missed by not holding those names came to -0.40 % a year (negative means the unavailable names went the wrong way for a short, so missing them helped). The borrow-aware optimiser, allowed to substitute from the next decile, brings fees to 0.67 % and recalls to 0.06 % (1.6 a month, locate failures 2.5 %): a recovery of 0.51 % of NAV a year in fees and recalls against the same optimiser without the borrow term (0.39 % all-in once the missed-shorts term is included), for a net return of 32.34 % against 30.92 % naive (1.42 %). Without any lending friction the same book returns 31.80 %.

MOM: the four books

book	gross price P0	long leg	short leg	shadow short	missed shorts	borrow fees	recalls	rebalance impact	rebate + cash	net return (lo	vol	max DD	fee on short l	locate failure	recalls / 100	missing short-
frictionless	42.30 %	38.88 %	3.42 %	3.42 %	0.00 %	0.00 %	0.00 %	1.69 %	1.91 %	31.80 %	46 %	~46 %	0.00 %	0.0 %	0.0	0.0 %
naive	42.56 %	38.79 %	3.77 %	3.37 %	-0.40 %	1.15 %	0.08 %	1.68 %	1.90 %	30.92 %	46 %	~45 %	1.37 %	3.6 %	3.4	3.8 %
risk_only	42.63 %	38.85 %	3.79 %	3.38 %	-0.41 %	1.16 %	0.08 %	1.67 %	1.90 %	30.96 %	46 %	~45 %	1.38 %	3.6 %	3.4	3.8 %
borrow_aware	43.33 %	38.89 %	4.44 %	4.15 %	-0.29 %	0.67 %	0.06 %	1.69 %	1.90 %	32.34 %	46 %	~45 %	0.92 %	2.5 %	2.7	2.6 %

All flows are annualised as a share of the running NAV; the net return is the annualised log return of the NAV. The shadow short leg holds every intended short without locates or recalls, so 'missed shorts' is the price return the real short leg did not get (the shadow leg's return less the real short leg's).

Optimiser (lambda = 20.0, alpha spread 1.0 % a month, tracking-variance penalty against the naive book): the risk-only book has an active share of 1.2 % against the naive deciles and the borrow-aware book 4.4 %; 99 % of the borrow-aware short leg stays in the bottom decile; the expected monthly borrow cost of the short leg falls from 22 bp to 12 bp of NAV.

The three books

book	signal	fee, naive	fee, aware	locate fail, n	aware	recalls / 100	fees + recalls	aware	missed shorts,	aware	net, naive	net, risk-only	net, borrow-aw	net, no fricti
MOM	12-1 momentum	1.37 %	0.92 %	3.6 %	2.5 %	3.4 / 2.7	1.23 %	0.73 %	-0.40 %	-0.29 %	30.92 %	30.96 %	32.34 %	31.80 %
SIR	short interest	4.95 %	2.59 %	5.7 %	2.7 %	6.7 / 4.8	3.52 %	2.02 %	-0.49 %	-1.15 %	-26.47 %	-26.66 %	-24.74 %	-23.33 %
REV	one-month reve	0.73 %	0.57 %	4.8 %	4.2 %	2.5 / 2.4	0.57 %	0.46 %	1.58 %	1.79 %	-11.13 %	-11.19 %	-11.44 %	-8.90 %

The borrow model

Fee source: quantile_map (no Interactive Brokers snapshots were reachable; see the data section). On 2026-09-17, 797 names: 85 % general collateral (fee under 1 %; the published figure is 85-92 %), 8.0 % special (over 5 %), mean fee 1.89 %, median 0.30 %, p90 3.42 %, p99 20.00 %; 12 names on a Reg SHO threshold list, 123 with utilisation above 80 % of modelled lendable supply, the utilisation floor binding for 101. Names on a threshold list carry a mean fee of 7.42 % against 1.15 % off it.

					fee buck
1-5%		1.32 %	1.2	87	30
>5%		1.48 %	1.0	87	12
gc		0.97 %	1.4	87	448

General collateral minus special: -0.51 % a month (t = -0.5, 87 months). The literature's sign is positive (expensive-to-short names underperform); it is a check on the ordering, not a strategy.

Sensitivity

fee scale	hazard scale	fees + recalls, naive	borrow-aware	all-in drag, naive	borrow-aware	recalls / month naive	aware	net naive	net aware	gain
0.5	1.0	0.66 %	0.47 %	0.26 %	0.95 %	1.5	1.6	31.48 %	31.32 %	-0.16 %
1.0	1.0	1.23 %	0.73 %	0.83 %	0.44 %	1.5	1.6	30.92 %	32.34 %	1.42 %
1.5	1.0	1.81 %	0.98 %	1.41 %	1.38 %	1.5	1.8	30.35 %	31.60 %	1.25 %
1.0	0.5	1.18 %	0.72 %	0.26 %	0.67 %	0.8	0.9	31.51 %	32.01 %	0.49 %
1.0	1.5	1.31 %	0.77 %	0.83 %	0.67 %	2.1	2.6	30.97 %	32.05 %	1.08 %
1.5	1.5	1.91 %	1.04 %	1.44 %	1.46 %	2.1	2.6	30.38 %	31.48 %	1.10 %
2.0	2.0	2.32 %	1.30 %	2.29 %	1.51 %	2.8	3.7	29.63 %	31.77 %	2.14 %

seed	fees + recalls, naive	borrow-aware	missed shorts, naive	locate failures, naive	recalls / month	net naive	net aware
1	1.23 %	0.73 %	-0.40 %	3.6 %	1.5	30.92 %	32.34 %
2	1.23 %	0.71 %	-0.75 %	4.1 %	1.4	31.30 %	32.66 %
3	1.21 %	0.75 %	-1.76 %	3.8 %	1.7	32.44 %	32.42 %

alpha spread / month	lambda	shorts held, aware	fees + recalls, naive	borrow-aware	fees aware	locate fail aware	net naive	net aware
0.5 %	20.0	71	1.23 %	0.75 %	0.67 %	2.3 %	30.92 %	32.41 %
2.0 %	20.0	53	1.23 %	0.74 %	0.67 %	3.0 %	30.92 %	30.86 %
1.0 %	5.0	53	1.23 %	0.54 %	0.48 %	2.3 %	30.92 %	33.45 %
1.0 %	80.0	64	1.23 %	0.97 %	0.90 %	2.6 %	30.92 %	31.52 %

Financing and the prime-broker reconciliation

On the naive MOM book the short rebate earns 1.91 % of NAV a year (EFFR on the short proceeds, ACT/360), borrow fees cost 1.15 % and the cash balance -0.01 %: net financing 0.75 % a year over 1815 days and 163,563 accrual lines. On the last day the book is \$8,510m gross (1.84x NAV): Regulation T initial margin \$4,255m (excess \$380m), maintenance \$2,936m, portfolio margin \$1,276m (excess \$3,358m). Reg T maintenance was breached on 1 days and portfolio margin on 0 days; the minimum excess equity was -9 % of NAV under Reg T and 41 % under portfolio margin.

The simulated PB statement has 163,488 lines against 163,563 in the book (163,589 date-symbol-kind keys). The reconciliation raised 2,930 breaks (2,822 above the \$0.50 tolerance) and 0 on keys with nothing seeded; 99.6 % of the seeded discrepancies were found and classified as the right type.

					seeded
stale_fee_rate	rate_break	1069	1061	0	
rebate_act365	day_count_break	42	42	0	
missing_rebate	missing_in_pb	242	242	0	
duplicate_fee	duplicate_line	161	161	0	
collateral_102	collateral_basis	957	957	0	
stale_mark	notional_break	388	385	3	
old_symbol	symbol_mapping	20	20	0	
buy_in_charge	unknown_charge	6	6	0	
stale_effr	rate_break	53	53	0	

break type	keys	net \$ (PB minus book)
rate_break	1114	76,158
collateral_basis	960	-15,786
notional_break	385	-3,816
missing_in_pb	242	998,894
duplicate_line	161	-288,585
day_count_break	42	-6,498
symbol_mapping	20	54,666
unknown_charge	6	-357,749

Data

As of 2026-09-17: 801 names, 1,257,350 name-days in the borrow table from 2019-06-03; store: 1,576,760 price bars, 121,021 FINRA short-interest rows over 184 settlement dates, 5,512 threshold-list rows over 1760 days, 609,820 fails-to-deliver rows, 60 ticker changes in the point-in-time map, 0 IB shortable rows, 2941 EFFR days. Runtime 10 minutes.